AS3: asymmetric initial lanes, useful finite advice changes and the complete gradient reference

Elementary VSL lane-drop task review

19 October 2026

Status. AS1 and AS2 are user-accepted. This completes AS3 at review/reference scope and returns to the original Step 3 user closure gate. No traffic rollout, numerical derivative, AD or optimization is run. No new traffic law or numerical smoothing is selected.

Result and limitation. The accepted hidden A/B split can matter for complete-TST advice, not merely for local receiving: a conditional existence range of the selected model has opposite utility for the same finite slowdown. A nonzero common-fastest range also exists. The positive range may require large forced work, weak mismatch and fast recovery; it is not an admitted physical parameter range. The complete regular-event sensitivity below is a reference for later verification, not proof of numerical gradient accuracy or usefulness.

1 Accepted traffic and information remain fixed

Use the accepted AS1 forward contract and accepted AS2 observation contract. E ends, A is its adjacent continuing lane and B the far lane. The initial uniform footprint is $[0, \ell]$, $L = L_p + b$ and

$$e_0, a_0, b_0 \geq 0, \quad r = e_0 + a_0, \quad \rho = e_0 + a_0 + b_0, \quad M = \ell\rho. \tag{1}$$

Each true initial state is fixed as advice $v \in [v_-, v_+]$ varies. So are geometry, physical coefficients, the W-F anchor and $m = 1$ veh-eq. Retain independent lane loads, neutral initial terminal/recovery histories, no postdecision arrivals, finite storage and choke completion. In particular,

$$v_+ r < R \leq C(v_-), \quad v_+ b_0 \leq R_B \leq C(v_-), \quad M < k_j b, \quad c_0 - \beta(1 + v_+/w) > 0, \tag{2}$$

with all other AS1 bounds intact. Write $K = wk_j$, $C(v) = Kv/(w + v)$, $k_c(v) = K/(w + v)$, $U(g) = wg/(K - g)$ and $c_0 = 1 + \delta \geq 1$, $\chi = \beta/[C(v_+)v_+]$ fixed once.

Level 2 receives E and continuing totals and every permitted history/inference; Level 3 additionally knows A/B separately. The compact peer knows one extra initial adjacent mass and exactly matches Level 3. No realized r, future E/service trace or pair label is leaked into Level 2. AS2's stronger matched aggregate/E history is retained, and identifying histories remain admissible.

2 A genuinely asymmetric analytical reference family

2.1 The same E and continuing total, with unequal adjacent loading

For arbitrary positive scales u, κ, h, choose

$$\begin{gathered} v_- = u,\ v_+ = 2u,\ w = 2u,\ k_j = 3\kappa,\ \gamma = 2, \quad R = 39\kappa u/20,\ R_B = \kappa u, \\ \nu = \ell/h = \log(6/5), \quad L_p = 2\ell, \quad b = \ell, \quad L = 3\ell, \\ \mathcal{L} : (e_0, a_0, b_0) = (3\kappa/5, \kappa/5, 7\kappa/20), \\ \mathcal{H} : (e_0, a_0, b_0) = (3\kappa/5, 7\kappa/20, \kappa/5). \end{gathered} \tag{3}$$

Here $\rho = 23\kappa/20$ is common, $r_{\mathcal{L}} = 4\kappa/5$, $r_{\mathcal{H}} = 19\kappa/20$, and $M/(k_j b) = 23/60 < 1$. The worst ordinary load margin is $R - v_+ r_{\mathcal{H}} = \kappa u/20$. AS2's same lower-speed loading construction reaches both initial states with identical full permitted histories, including E history.

The family fixes initial state/geometry; it does not fix one work cost for all deductions. AS2 illustrated $c_0 = 11/10, \beta = 1/100$. The action existence proof below chooses another *shared* work range inside the accepted domain, not a new result for those illustrative coefficients. Predecision histories stay unchanged since no traffic reached the terminal.

2.2 Preparation and its nonzero asymmetric speed sensitivity

With $z = vt$ and $\Delta_i(v) = k_c(v) - r_i > 0$, H1 is

$$e_{i,z} = -\frac{1}{h}\min\{e_i, (\Delta_i + e_i)/2\}. \tag{4}$$

$\mathcal{L}$ is donor-limited at every advice: $\Delta_{\mathcal{L}} \geq 7\kappa/10 > e_0$. Its density at every fixed progress is independent of advice. $\mathcal{H}$ starts receiving-limited only when $v > v_c = 58u/31$. Its switch progress is

$$z_s(v) = 2h \log \frac{e_0 + \Delta_{\mathcal{H}}(v)}{2\Delta_{\mathcal{H}}(v)}. \tag{5}$$

At its largest value, $v = 2u$, $\exp(z_s/h) = (23/22)^2 < 6/5 = \exp(\ell/h)$. Thus all terminal handoff exposures $z \in [\ell, 2\ell]$ are donor-branch exposures, with front density $B_i(v) = e_i(\ell; v)$:

$$B_{\mathcal{L}} = \frac{\kappa}{2}, \qquad B_{\mathcal{H}}(v) = \begin{cases} \kappa/2, & v \leq 58u/31, \\ \dfrac{5}{6}\dfrac{(e_0 + \Delta_{\mathcal{H}})^2}{4\Delta_{\mathcal{H}}}, & v > 58u/31. \end{cases} \tag{6}$$

The two expressions match with zero first derivative at v_c. At v_+,

$$B_{\mathcal{H}} = \frac{529\kappa}{1056}, \quad B'_{\mathcal{H}} = \frac{115\kappa}{7744u} > 0, \qquad B_{\mathcal{L}} = \frac{\kappa}{2}, \quad B'_{\mathcal{L}} = 0. \tag{7}$$

This replaces the historical symmetric-member calculation; its different E population is not reused as the present hidden variable.

3 A nonzero common-fastest range in the selected law

The front free-work test is $W_{f,i}(v) = v[r_i + \delta B_i(v)]$. H1 comparison makes B_i nondecreasing in advice, so the largest front work is at $2u$. There,

$$\frac{W_{f,\mathcal{L}}}{\kappa u} = \frac{8}{5} + \delta, \qquad \frac{W_{f,\mathcal{H}}}{\kappa u} = \frac{19}{10} + \frac{529}{528}\delta. \tag{8}$$

Consequently, for

$$0 \leq \delta \leq \frac{132}{2645}, \qquad J_i(v) = \frac{M(L - \ell/2)}{v} \quad \text{for both states and every advice.} \tag{9}$$

Joint free admission holds; its workload decreases through the pulse. Free E's mark is $v \geq U(R)$, so no recovery loss is deposited. At equality the selected free-continuation guard applies. This includes positive maneuver burden and can have positive β and finite recovery acceleration. Both preparations uniquely prefer v_+. Information can be missing without being needed for the best action in this range.

4 Complete TST can have different useful advice changes

4.1 An exact auxiliary reference, with every waiting tail counted

For this subsection only, delete mismatch and retained recovery while keeping H1, propagation, c_0, initial state and TST. Denote this fixed-work reference by J_i^0. A remains free because $\lambda_A < R$; while E queues, $g_E = (R - \lambda_A)/c_0$. This is an auxiliary reference/rival, not an amendment of the selected law.

Let $t_f = (L - \ell)/v$, $d = \ell/v$, and $s = t - t_f$ during the pulse. Since every exposure is past the applicable preparation switch,

$$\lambda_E = vB_i(v)e^{-vs/h}, \quad \lambda_A = v[r_i - B_i(v)e^{-vs/h}], \quad 0 < s < d. \tag{10}$$

Set

$$m_i = hB_i(v)(1 - e^{-\nu}) = hB_i(v)/6, \quad Q_i = hB_i(v)\ell - hm_i, \quad D_i = \ell r_i - Rd, \quad \mathcal{T} = M(L - \ell/2). \tag{11}$$

m_i is forced E mass, distinct from the fixed renewal mass m; Q_i has units mass times length and is an arrival-pulse area quantity. No normalization or packet size is changed.

A sufficient condition for a strictly positive E queue throughout the eligible pulse at every advice is

$$c_0 > 94/25. \tag{12}$$

Indeed e_i at any handoff exposure is at least $e_0 e^{-2\nu} = 5\kappa/12$, $r_i \geq 4\kappa/5$, $v \geq u$, and $u[4\kappa/5 + (c_0 - 1)5\kappa/12] > R$. The queue is

$$q_i^0(s) = \frac{(vr_i - R)s + (c_0 - 1)hB_i(1 - e^{-vs/h})}{c_0}. \tag{13}$$

It drains after the rear at R/c_0. Integrating the pulse and full tail gives

$$J_i^0(v) = \frac{\mathcal{T}}{v} + \frac{D_i d}{2c_0} + (1 - 1/c_0)\frac{Q_i}{v} + \frac{[D_i + (c_0 - 1)m_i]^2}{2c_0 R}. \tag{14}$$

Every initial lane's travel and the entire E tail are included; A has no waiting in this reference. Equation (14) is used only on its proved branch.

4.2 The reference action contrast is proved anew for fixed E

For $\mathcal{L}$, B, m_i, Q_i are action-independent. Writing $W_i = \ell r_i + (c_0 - 1)m_i$, exact cancellation of the v^{-2} terms gives

$$J_{\mathcal{L}}^0(v) = \frac{W_{\mathcal{L}}^2}{2c_0 R} + \frac{A_{\mathcal{L}}}{v}, \quad A_{\mathcal{L}} = \mathcal{T} + (1 - 1/c_0)(Q_{\mathcal{L}} - \ell m_{\mathcal{L}}) - \frac{\ell^2 r_{\mathcal{L}}}{2c_0} > 0. \tag{15}$$

To see strictness, $0 \leq Q_i \leq \ell m_i$, $m_i \leq \ell r_i$ and $\mathcal{T} = (5/2)\ell M > \ell^2 r_i$ imply $A_{\mathcal{L}} \geq \mathcal{T} - \ell^2 r_{\mathcal{L}} > 0$. Hence v_+ is the unique optimum of the low-A reference.

For $\mathcal{H}$, expand the same complete formula as

$$\begin{aligned} J_i^0 &= \frac{c_0 m_i^2}{2R} + I_{0,i} + I_{1,i}/c_0, \\ I_{0,i} = \frac{\mathcal{T} + Q_i}{v} + \frac{(D_i - m_i)m_i}{R}, &\quad I_{1,i} = \frac{D_i d}{2} - \frac{Q_i}{v} + \frac{(D_i - m_i)^2}{2R}. \end{aligned} \tag{16}$$

At v_+, $A_+ = m_{\mathcal{H}} m'_{\mathcal{H}}/R > 0$, whereas $I'_{0,\mathcal{H}}$ and $I'_{1,\mathcal{H}}$ are finite and independent of c_0. Thus

$$c_0 > \max\left\{94/25, \frac{|I'_{0,\mathcal{H}}(v_+)| + |I'_{1,\mathcal{H}}(v_+)|}{A_+}\right\} \quad \Longrightarrow \quad (J_{\mathcal{H}}^0)'(v_+) > 0. \tag{17}$$

Near v_- the high-A preparation is donor-only, and the same positive A_i/v coefficient gives a strictly negative derivative. The continuous reference therefore has interior high-A minimizers and disjoint low/high minimizing sets. A sufficiently small common feasible slowdown from v_+ strictly worsens $\mathcal{L}$ and improves $\mathcal{H}$. No physical plausibility or moderate-work result follows from this existence bound.

4.3 Transfer finite action utility into the selected positive-mechanism law

Restore the actual fixed W-F mismatch, E-only source and recent-mass O-R renewal. With $u_R = U(R)$, its zero-mark isolated loss area and account satisfy

$$W_0 = \frac{K}{w + u_R} \frac{u_R^2}{2a_r}, \qquad 0 \le p(t) \le \varepsilon_p := \frac{RW_0}{mc_0}. \tag{18}$$

The latter bounds the nonnegative retained-window time integral by the all-history integral with $g_E \le R/c_0$. It does not replace renewal. Put $a_* = v_+ \max_i r_i < R$ and choose

$$0 < \beta \le 1, \quad c_0 - \beta(1 + v_+/w) > 0, \quad \varepsilon_p < \min\{(R - a_*)/2, R/2\}. \tag{19}$$

Then $\mathcal{B} \ge R - \varepsilon_p > a_*$, so A is free by its actual admission test. While E queues, its service is bracketed by

$$\mu_- := \frac{R - \varepsilon_p - \lambda_A}{c_0 + \beta} \le g_E \le \mu_0 := \frac{R - \lambda_A}{c_0}, \quad 0 \le \mu_0 - \mu_- \le \varepsilon_\mu := \frac{\varepsilon_p}{c_0} + \frac{R\beta}{c_0^2}. \tag{20}$$

The same arrivals drive reflected comparison queues q_0, q_- with these two positive services. The causal free/queued guards and a first-crossing comparison give $q_0 \le q_E \le q_-$; the actual queue itself need not be a fixed-service reflection. Reflection implies $q_- - q_0 \le t\varepsilon_\mu$. Choose $H > L/v_- + 2(c_0 + 1)M/R$ after fixing c_0 and geometry. Every compared cohort clears by H. Consequently, for both states and all advice,

$$0 \le J_i(v) - J_i^0(v) \le \varepsilon_J := H^2\varepsilon_\mu/2. \tag{21}$$

The initial split is held fixed in this estimate. It makes no derivative closeness assertion. With c_0 fixed first, positive β can be made small and positive finite a_r sufficiently large to make ε_J as small as needed while retaining every selected mechanism.

Let $d_i^0(v) = J_i^0(v) - \min_z J_i^0(z)$. Compactness, continuity and disjoint reference minimizing sets give

$$\Gamma_0 = \min_v \max_i d_i^0(v) > 0. \tag{22}$$

For actual $V_i = \inf_v J_i(v)$, uniform comparison yields conservatively

$$\inf_v \max_i \{J_i(v) - V_i\} \ge \Gamma_0 - 2\varepsilon_J > 0 \qquad (\varepsilon_J < \Gamma_0/4). \tag{23}$$

No global continuity or attainment of actual minima is assumed. Making $2\varepsilon_J$ smaller than each of the two strict reference slowdown gaps also preserves their opposite signs in the selected law. Thus identical Level 2 histories can require incompatible near-optimal common advice. Level 3 and its compact peer can distinguish the states.

Limits. This is a new asymmetric action certificate, not the old hidden-E result renamed. It may need large work, weak mismatch and fast recovery. A stays free in this range; the archived A-delay/active-retirement witness is separate. No joint strong-delay/action/physical witness, default calibration, opposite infinitesimal gradient signs in the actual law, AD advantage or real-freeway usefulness is proved.

5 The complete sensitivity reference does not require initial symmetry

5.1 Preparation, clocks and all three lane arrivals

Define $E_v(z;v) = \partial_v e(z;v)$ at fixed physical progress, with $E_v(0;v) = 0$. On each smooth H1 branch,

$$(E_v)_z = \begin{cases} -E_v/h, & \text{donor}, \\ -(E_v + k_c'(v))/(\gamma h), & \text{receiving}, \end{cases} \qquad k_c'(v) = -K/(w+v)^2. \tag{24}$$

The vector fields match at their moving switch, so the state sensitivity has no jump from unequal vector fields there. Every true r, b_0, e_0 is fixed within the advice derivative; one does not regenerate initial traffic or recondition its history after changing advice.

For an initial position x, eligibility is $(L-x)/v$. Exactly,

$$P_E(t;v) = \int_0^\ell e(L_p - x; v)\mathbf{1}_{(L-x)/v \le t}\,\mathrm{d}x. \tag{25}$$

Inside the eligible pulse set $x_* = L - vt$, $z_* = vt - b$. Then

$$\begin{aligned} P_{E,v} &= \int_{x_*}^\ell E_v(L_p - x; v)\,\mathrm{d}x + te(z_*; v), \\ \lambda_E &= ve(z_*; v), \qquad \lambda_{E,v} = e + vE_v + vte_z, \\ P_{A,v} &= rt - P_{E,v}, \quad P_{B,v} = b_0 t, \quad \lambda_{A,v} = r - \lambda_{E,v}, \quad \lambda_{B,v} = b_0, \\ t_f' &= -(L-\ell)/v^2, \qquad t_r' = -L/v^2. \end{aligned} \tag{26}$$

Before the front all cumulative sensitivities vanish; after the rear the E derivative is its full progress integral, A is its negative and B is zero. Their final total sensitivity is zero by fixed total mass. Use one-sided front/rear values; fixed clocks would omit a genuine causal channel.

5.2 Implicit point service and its traffic marks

For queued E retain $\mathcal{H}(g;a,u) = g[c_0 + \chi a[u - U(g)]_+] = x$. Here a is A flow. On positive mismatch,

$$\begin{gathered} \mathcal{H}_g = c_0 + \chi a\{u - U(g) - gU'(g)\} > 0, \quad \mathcal{H}_a = \chi g\{u - U(g)\}, \quad \mathcal{H}_u = \chi a g, \\ g_v = \frac{x_v - \mathcal{H}_a a_v - \mathcal{H}_u u_v}{\mathcal{H}_g}. \end{gathered} \tag{27}$$

On inactive mismatch, $\mathcal{H}_g = c_0$, $\mathcal{H}_a = \mathcal{H}_u = 0$. Free A/queued E uses $(a,u,x) = (\lambda_A, v, \mathcal{B} - \lambda_A)$; both queued use $(\mathcal{B}/2, U(\mathcal{B}/2), \mathcal{B}/2)$. Free marks have derivative one; queued marks have derivative $U'(g)g_v$. The other service branches are differentiated from their retained admission equations. At fixed current traffic/budget the fully queued root has no direct advice term. χ is not recomputed with advice or a search box.

5.3 History, retirement and queue events

Write $Y = Y_E$, $g = g_E$ and $Z = Y_v$. On a regular history chart,

$$p(t) = \frac{1}{m}\int_{s_m(t)}^t D(t-s; u_E(s))g_E(s)\,\mathrm{d}s, \quad Y(s_m) = Y(t) - m \quad \text{after fill}, \tag{28}$$

with $s_m = 0$ before fill. After fill suppose the cutoff lies at positive old E rate, away from a mark event. Then $g(s_m)s_{m,v} = Z(t) - Z(s_m)$. For source/mark events $\tau_k(v)$ inside the retained

window put $\Phi_k^{\pm} = D(t-\tau_k; u_E(\tau_k^{\pm}))g_E(\tau_k^{\pm})$ and $D_o = D(t-s_m; u_E(s_m))$. Piecewise Leibniz differentiation gives

$$p_v = \frac{1}{m}\Big[\int_{s_m}^{t}(D_u u_{E,v} g_E + D g_{E,v})\,\mathrm{d}s + \sum_k (\Phi_k^- - \Phi_k^+)\tau_k' \\ - \mathbf{1}_{Y(t)>m} D_o\{Z(t) - Z(s_m)\}\Big], \qquad \mathcal{B}_v = -p_v. \tag{29}$$

The last term is old-weight retirement; the old-rate denominator cancels. It must not be omitted or replaced by a strictest-old-cap rule. For active loss, with $S = \sqrt{(w+u)^2 + 2wa_r\theta}$,

$$D_u = -Kw(w+u)/S^3, \qquad D_\theta = -Kw^2 a_r/S^3. \tag{30}$$

They are zero on the inactive side. At fixed integration time s, physical age $t-s$ is fixed; moving source-event times enter through the jump terms. The zero-loss support edge has no jump contribution because D is continuous and zero there. Fill, plateau and cutoff/mark encounters require their own one-sided/event treatment, not a universal smooth derivative assertion.

For an isolated transverse guard let $G(t,v)$ denote its left-side value evaluated along the full history-aware solution. Then

$$\tau_k' = -G_v^-/G_t^-, \qquad X_v^+ = X_v^- + (F^- - F^+)\tau_k' \tag{31}$$

for continuous physical-state coordinates with identity reset. Guard derivatives include the history sensitivity above. This notation does not claim a finite-dimensional Markov reduction of the account. A birth uses the free-feasibility margin $\mathcal{B} - \lambda_A$, not the identically zero free A queue; clearance uses the positive queue approaching zero. Recorded source-event times remain in the history. The regular timing principle is the one reviewed in Kong et al.; the renewal derivative is a separate model-specific deduction.

5.4 Complete TST and the derivative's domain

Use a fixed horizon above the AS1 clearance bound, independent of advice:

$$H_* > L/v_- + c_{\max}M/B_{\min}, \qquad B_{\min} > 0.$$

With $Z_j = Y_{j,v}$,

$$J = \int_0^{H_*}(M - Y_A - Y_E - Y_B)\,\mathrm{d}t = \frac{M(L-\ell/2)}{v} + \int_0^{H_*}(q_A + q_E)\,\mathrm{d}t, \\ J' = -\int_0^{H_*}(Z_A + Z_E + Z_B)\,\mathrm{d}t = -\frac{M(L-\ell/2)}{v^2} + \int_0^{H_*}(q_{A,v} + q_{E,v})\,\mathrm{d}t. \tag{32}$$

Queue-area endpoint values vanish at zero-queue hits; their timing effects on future service/history do not. All traffic, including both A/E waiting tails, is counted. Recovery after the last completed unit is not extra own travel beyond the selected choke completion.

On a neighborhood with a fixed finite event list, smooth branches, transverse guards and regular cutoffs, differentiating the locally uniform causal Volterra contraction gives the within-chart linear sensitivity. Implicit guard differentiation and identity-state matching join the charts; bounded regular-chart terms permit fixed-horizon differentiation. This proves the local reference construction. It uses no initial A/B equality.

Forward-admitted grazing, coincident events, activation thresholds, root/ profile kinks and renewal encounters remain in the task. Check continuity and feasible side behavior before assigning a derivative. If no finite ordinary or one-sided derivative exists, report that and the actual finite cost change. No smoothing, frozen event, deleted boundary case or new generalized-gradient controller is selected. Uniform value comparison does not imply gradient closeness or global minimum attainment.

6 Evidence and source limits at the original Step 3 gate

AS1's complete-forward result, AS2's conditional information contrast and compact equality, and this asymmetric action/sensitivity review now fit one standalone verification contract. Later evidence must distinguish continuous reference correctness, numerical forward convergence, numerical differences, AD, feasible-step usefulness and optimizer performance. A strong bounded scalar search and the compact peer remain required rivals. No such numerical layer is marked passed here.

Information revealed through complicated features may be harder for a learner to extract; that is a plausible separate conditioning/learning hypothesis. AS2's exactly identical histories establish structural ambiguity, not that learning hypothesis. When information is recoverable from the same inputs, model-aware inference and feature transforms must be allowed equally; learning difficulty needs matched tasks, inputs and resources before a claim. No training experiment is opened by this clarification.

This note adapts the own fixed-work derivation and service/queue comparison in the historical standalone Step 3 mathematics, with the new fixed-E split calculation and explicit sensitivity audit above. It does not transplant the historical symmetric action certificate. The archived Kong et al. hybrid-sensitivity paper supports regular event-time corrections only; the Delle Monache–Piccoli–Rossi VSL paper is prior art under a different objective. No paper is newly viewed or new traffic model adopted in AS3. Applied MP6 remains unexecuted, SUMO optional, Step 4/implementation unopened and ramp/urban lines closed.